W. M. Kalani Sriya Ilmini, Ph.D.

Senior Lecturer in Computer Science • AI Researcher

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Academic Profile

Senior Lecturer in Computer Science with teaching, research, supervision and academic leadership experience across undergraduate and postgraduate programmes. Research work spans machine learning, deep learning, explainable artificial intelligence, computer vision, biomedical and healthcare applications, knowledge-based systems, and emerging work on large language models and agent systems. Recipient of the Gold Medal for Best Postgraduate Student in the Physical Science Stream at the 50th General Convocation of the University of Sri Jayewardenepura. Active in international research networking through COST Action CA24121 on Knowledge Graphs and Large Language Models.

Machine Learning | Deep Learning | Explainable AI | Biomedical AI | Knowledge Graphs | Large Language Models

Academic Appointments

2025–Present — **Senior Lecturer Grade II**, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, Sri Lanka.

2024–2025 — **Senior Lecturer Grade II – On Contract**, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, Sri Lanka.

2025–2026 — **Visiting Lecturer**, NIBM, Sri Lanka.Advanced Diploma in Data Science

2024 — **Visiting Lecturer**, Next Campus, Sri Lanka and Dubai (in-person and online).

Delivered lectures to Sri Lankan and Dubai student cohorts in hybrid formats and designed course content for diverse international audiences.

2023–2024 — **Senior Lecturer Grade II**, Department of Computer Science, Faculty of Computing, General Sir John Kotelawala Defence University, Sri Lanka.

2017–2023 — **Lecturer**, Department of Computer Science, Faculty of Computing, General Sir John Kotelawala Defence University, Sri Lanka.

2014–2016 — **Temporary Lecturer**, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, Sri Lanka.

2015 — **Visiting Instructor, MSc in Computer Science**, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, Sri Lanka.

2014 — **Temporary Instructor**, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, Sri Lanka.

2013–2014 — **Software Developer (Intern)**, Solution Infinity (Private) Limited, Sri Lanka.

Education

2017–2023 **Ph.D.**, University of Sri Jayewardenepura

Thesis: *Personality Traits Assessment using Facial Features with Deep Learning Techniques*.

2008–2014 **BSc Honours Special in Computer Science**, University of Sri Jayewardenepura

Department of Computer Science, Faculty of Applied Sciences. GPA: 3.7/4.5.

Awards and Distinctions

2024 **Gold Medal – Best Postgraduate Student, Physical Science Stream**

50th General Convocation, University of Sri Jayewardenepura, Sri Lanka.

2023 **Best Poster Award, Computing Track – 16th International Research Conference KDU**

NTD Dharmasiri and WMKS Ilmini, *Home Garden-based Ayurvedic Plant Identification System Using Convolutional Neural Networks*.

2021 **Best Poster Award, Computing Track – 14th International Research Conference KDU**

RM Fernando, WMKS Ilmini, and DU Vidanagama, *Air Quality Prediction Using Machine Learning*.

2020 **National Best Quality Software Award – Tertiary (Undergraduate) Merit Award**

Mobile App for Automatic Identification of Snake Types Found in Sri Lanka Using Convolutional Neural Networks, Savini Abayarathne, WMKS Ilmini and TGI Fernando.

2019 **National Best Quality Software Award – Tertiary (Postgraduate) Merit Award**

Personality Detector, WMKS Ilmini and TGI Fernando.

Teaching and Research Supervision

Postgraduate teaching Python for Data Science; Machine Learning I; Data Structures and Alogrithms, Bioinformatics

Undergraduate teaching Machine Learning; Machine Learning for Data Science; Programming for Data Science; Object-Oriented Programming; Automata Theory; Bioinformatics; Operational Research; Fundamentals of Programming; Social Aspects of Computing; Software Architecture and Design; Data Structures and Algorithms.

Supervision & coordination Supervision of numerous BSc, MSc and Mphil research projects in ML, DL and AI across different domains; final-year project coordination; curriculum development committee participation.

Research Publications

Journal Articles

1. V. S. Athukorala and W. M. K. S. Ilmini, “Explainable AI for critical care: A systematic review of interpretable models for sepsis and ICU mortality prediction,” *BMC Medical Informatics and Decision Making*, vol. 26, no. 1, p. 64, 2026. doi:10.1186/s12911-026-03344-0.
2. W. Ilmini and T. Fernando, “Detection and explanation of apparent personality using deep learning: A short review of current approaches and future directions,” *Computing*, vol. 106, no. 1, pp. 275–294, 2024.
3. W. Ilmini and T. Fernando, “Exploring the impact of facial features on apparent personality traits detection using deep learning techniques,” *KDU Journal of Multidisciplinary Studies*, vol. 4, no. 2, 2024.
4. W. Ilmini and T. Fernando, “Explaining the outputs of convolutional neural network-recurrent neural network (CNN-RNN)

based apparent personality detection models using the class activation maps," *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 2, 2023.

5. W. Ilmini and T. Fernando, "Performance analysis of state-of-the-art deep learning models in the visual-based apparent personality detection," *Vidyodaya Journal of Science*, vol. 25, no. 2, 2022.
6. R. Fernando, W. Ilmini, and D. Vidanagama, "Prediction of air quality index in Colombo," *International Journal of Research in Computing*, vol. 1, 2021.
7. K. Ilmini and T. Fernando, "Persons' personality traits recognition using machine learning algorithms and image processing techniques," *Advances in Computer Science: an International Journal*, vol. 5, no. 1, Jan. 2016, No. 19.

Conference Proceedings

1. A. A. Abeysekara, G. A. I. Uwanthika, W. M. K. S. Ilmini, and G. R. N. N. Waidyarathna, "Ontology-based knowledge representation for recommending dietary guidelines to non alcoholic fatty liver disease patients," in *2025 9th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI)*, 2025, pp. 1–6. doi:10.1109/SLAAI-ICAI68534.2025.11318479.
2. G. Divyanjalee, K. Ilmini, and I. Uwanthika, "FITSTYLE: A GAN-based application revolutionizing online shopping by enhancing the virtual try-on experience," in *2025 International Research Conference on Smart Computing and Systems Engineering (SCSE)*, Apr. 2025, pp. 1–6. doi:10.1109/SCSE65633.2025.11030967.
3. C. Hapuarachchi, W. Ilmini, and B. Balasooriya, "Transforming banking with data lakehouse architecture: Overcoming integration challenges to enhance analytics and decision-making," 2025.
4. A. Nuwan, W. Ilmini, and N. Wedasinghe, "Optimizing schema and indexing strategies for large databases: A comparative study of SQL, NoSQL, and HTAP systems," 2025.
5. P. Wickramage, W. Ilmini, and G. Ganepola, "A review on AI techniques for identifying pest-related diseases in coconut leaves," 2025.
6. H. Divyanjalee, W. Ilmini, and G. Uwanthika, "Revolutionizing online shopping: A comprehensive review of an automated size recommendation and immersive 3D visualization system for virtual try-on enhancement," in *Proceedings of the 4th Student Symposium of Faculty of Computing, KDU*, 2024.
7. S. Serasinghe, G. Uwanthika, and W. Ilmini, "Comparative analysis of diagnostic methods for differentiating Hodgkin lymphoma and non-Hodgkin lymphoma," in *Proceedings of the 4th Student Symposium of Faculty of Computing, KDU*, 2024.
8. Y. Yakandawala, M. Madushanka, and W. Ilmini, "Explainable AI for transparent phishing email detection," in *2024 International Conference on Advances in Technology and Computing (ICATC)*, IEEE, 2024, pp. 1–6.
9. P. Muthumali, W. Ilmini, and G. Waidyarathna, "Nutricare: Optimizing personalized diet recommendation through a knowledge-based approach," in *Proceedings of the 16th International Research Conference KDU*, 2023.
10. N. Nanayakkara, W. Ilmini, and N. Jayathilake, "A review on wireless sensor networks and object detection methods in military applications," in *Proceedings of the 16th International Research Conference KDU*, 2023.
11. M. Priyankara, A. Gunasekara, and K. Ilmini, "Sign language translation techniques using artificial intelligence for the hearing impaired community in Sri Lanka: A review," in *Proceedings of the 2023 7th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI)*, IEEE, 2023, pp. 1–6.
12. A. Amarathunga, J. Fernando, and W. Ilmini, "Convolutional neural networks based face mask detection and automated door entry control system," in *Proceedings of the 15th International Research Conference KDU*, 2022.
13. W. Ilmini and T. Fernando, "Facial feature identification in the deep learning based apparent personality detection," in *Proceedings of the 2nd International Conference on Advanced Research in Computing (ICARC)*, IEEE, 2022.
14. T. Nadeeshana and W. Ilmini, "Simultaneous detection of COVID-19 and its pneumonia using multitask learning," in *Proceedings of the 15th International Research Conference KDU*, 2022.
15. R. Fernando, W. Ilmini, and D. Vidanagama, "Air quality prediction using machine learning," in *Proceedings of the 14th International Research Conference KDU*, 2021.
16. G. Gunawardana and W. Ilmini, "Prevention of cyber bullying using machine learning techniques," in *Proceedings of the 14th International Research Conference KDU*, 2021.
17. J. Hansika and W. Ilmini, "Disease identification in leafy vegetables using transfer learning," in *Proceedings of the 13th International Research Conference KDU*, 2020.
18. S. Kao and W. Ilmini, "Automated generation of Sinhala lyrics using recurrent neural networks," in *Proceedings of the 13th International Research Conference KDU*, 2020.
19. R. Ranasinghe and W. Ilmini, "Introducing a LSTM based flood forecasting model for the Nilwala river basin with a mobile application–a review," in *Proceedings of the 13th International Research Conference KDU*, 2020.
20. T. Thenabadu and W. Ilmini, "Diabetes prediction system using machine learning," in *Proceedings of the 13th International*

Research Conference KDU, 2020.

21. M. Wijethunga and W. Ilmini, "Air quality predicting system for Colombo city using machine learning approaches," in *Proceedings of the 13th International Research Conference KDU*, 2020.
22. S. Abayarathne, K. Ilmini, and T. Fernando, "Identification of snake species in Sri Lanka using convolutional neural networks," in *Proceedings of the International Conference in Artificial Intelligence (SLAAI)*, 2019.
23. S. Abayarathne, W. Ilmini, and T. Fernando, "Automated method to identify snake species in Sri Lanka, a review," in *Proceedings of the 12th International Research Conference KDU*, 2019.
24. M. Diunugala, W. Ilmini, and W. Pemarathne, "Identification of tomato plant diseases using convolutional neural networks," in *Proceedings of the 12th International Research Conference KDU*, 2019.
25. M. Guruge, D. Kulasekara, and W. Ilmini, "A review on graphical symbols identification on food packages using machine learning," in *Proceedings of the 12th International Research Conference KDU*, 2019.
26. K. Ilmini and T. Fernando, "IPersonalityReader: An application to assess the personality of a person via face image using convolutional neural networks," in *Proceedings of the International Conference in Artificial Intelligence (SLAAI)*, 2019.
27. F. Sourjah, W. Ilmini, and W. Pemarathne, "Ayurvedic plant identification using transfer learning," in *Proceedings of the 12th International Research Conference KDU*, 2019.
28. P. Vinodi, W. Pemarathne, W. Ilmini, and G. Senananyake, "Breast cancer detection using convolutional neural networks," in *Proceedings of the 12th International Research Conference KDU*, 2019.
29. L. Wijesinghe, D. Kulasekara, and W. Ilmini, "An intelligent approach to segmentation and classification of common skin diseases in Sri Lanka," in *Proceedings of the National Information Technology Conference (NITC)*, 2019.
30. L. Wijesinghe, D. Kulasekara, and W. Ilmini, "Research directions for skin disease identification using image processing and machine learning," in *Proceedings of the 12th International Research Conference KDU*, 2019.
31. C. Wijewantha, W. Pemarathne, and W. Ilmini, "Reducing frequently killing of elephants from train collision using machine learning," in *Proceedings of the 12th International Research Conference KDU*, 2019.
32. D. Gamage and K. Ilmini, "Flood forecasting using artificial neural network for Kalu Ganga," in *Proceedings of the SLAAI International Conference*, 2018.
33. S. Upsehala and W. Ilmini, "Location based tourist guiding application for tourist in Sri Lanka using image processing," in *Proceedings of the 7th ICADA Conference on Advancement of Development Administration*, 2018.
34. K. Ilmini and T. Fernando, "Computational personality traits assessment: A review," in *Proceedings of the IEEE International Conference on Industrial and Information Systems (ICIIS)*, 2017.

Preprints

1. R. Aratchige and W. Ilmini, *LLMs Working in Harmony: A Survey on the Technological Aspects of Building Effective LLM-Based Multi-Agent Systems*, 2025.

International Research Networks and Professional Memberships

2026–Present — **Working Group Member – COST Action CA24121: Knowledge Graphs and Large Language Models (KGELL)**

European Cooperation in Science and Technology (COST), European Union. Working Group 1 (WG1): Augmenting Large Language Models with Knowledge Graphs.

2017–Present — **Professional Member**, IEEE (Institute of Electrical and Electronics Engineers).

2025–Present — **Life Member**, Sri Lanka Association for Artificial Intelligence (SLAAI).

2017–2025 — **Member**, Sri Lanka Association for Artificial Intelligence (SLAAI).

2018–2019 — **Professional Member**, Computer Society of Sri Lanka (CSSL).

2021–2023 — **Professional Member**, BCS, The Chartered Institute for IT.

Academic Leadership, Service and Coordination

2026–Present	**Senior Treasurer**, Computer Science Student Association, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura.
2024	**Head of the Department**, Department of Computer Science, Faculty of Computing, General Sir John Kotelawala Defence University.
2023–2024	**Chair**, 4th Student Symposium, Faculty of Computing, KDU.
2023	**Faculty Coordinator**, 16th International Research Conference, Faculty of Computing, KDU.
2017–2024	**Level Coordinator**, Level Three BSc in Computer Science, Department of Computer Science, KDU.
2017–2022	**International/Local Competition Coordinator**, Faculty of Computing, KDU.
2017–2024	**Learning Management System Coordinator**, Department of Computer Science, KDU.
2024	**Counsellor**, IEEE Student Branch, KDU.
2018–2024	**Academic Advisor**, IEEE Student Branch, KDU.
2024	**Member**, Curriculum Development and Review and Quality Assurance Committee, Faculty of Computing, KDU.
2017–2024	**Review Committee & Organizing Committee Member**, International Research Conference of KDU.
2022–2025	**Review Committee Member**, Journal of Multidisciplinary Studies, International Journal of Research in Computing, Faculty of Computing KDU, and other international journals such as Clinical Interventions in Aging By Dove Medical Press, Taylor & Francis Group and Frontiers in Digital Health
	Review Committee Member, KDU-IRC 2017–2026; SLAAI International Conference 2023–2026; ICAC 2023; ICARC 2023; and other international journals.
2019–2023	**Resource Person**, KDU Certificate in Teaching and Higher Education (CTHE) Program.
2020–2024	**Coordinator**, Intelligent Research Laboratory, Faculty of Computing, KDU.
2018–2024	**Final Year Project Coordinator**, BSc Hons in Computer Science and Software Engineering, KDU.
2022–2024	**Department Research Coordinator**, Department of Computer Science, KDU.